ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · B2 · SOIL

# Soil layers and soil types

★☆☆
Grade 5

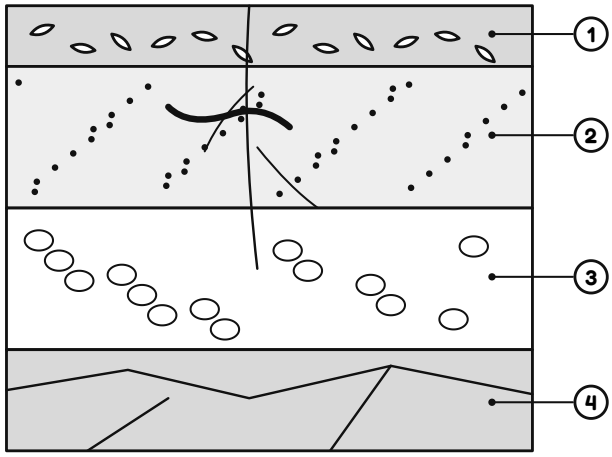
**Learning goals**

- Label soil layers.
- Use a key to identify soils.

**1** **Label the layers of soil.** [4]

LABELS

**subsoil** **leaf litter (humus)** **bedrock** **topsoil**



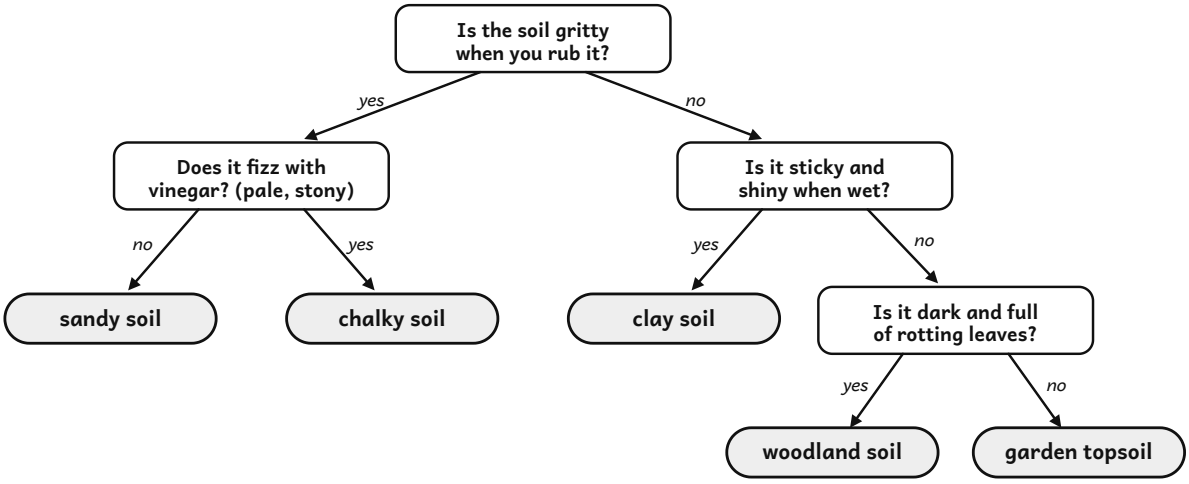

1 leaf litter (humus)

2 topsoil

3 subsoil

4 bedrock

**2** **Use the key to identify each soil sample.** [5]




| Sample | What the class observed | Soil type |
|---|---|---|
| A | gritty, does not fizz | sandy soil |
| B | not gritty, sticky and shiny when wet | clay soil |
| C | not gritty, not sticky, dark with rotting leaves | woodland soil |
| D | gritty, pale and stony, fizzes | chalky soil |
| E | not gritty, not sticky, dark and crumbly, few leaves | garden topsoil |

**3** **Fill in the gaps.** [5]

WORD BANK

**plastic** **air** **dry** **nutrients** **skin** **moist** **humus**

Soil is a mixture of rock particles, humus, water and air. Earthworms need moist soil because they breathe through their skin. Plants grow best in moist soil with plenty of nutrients.

ANSWER KEY

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LEARN · PLAY · WONDER

THINK IT THROUGH · B2 · SOIL

# Soil, worms and plants

★★☆
Grade 5

**Learning goals**

- Explain soil properties.
- Link soil to plant growth and soil animals.

**1** **Why does water drain quickly through sandy soil but slowly through clay?** [2]

Sand particles are large with big gaps between them, so water runs through; clay particles are tiny and packed tightly, so water moves slowly and is held.

**2** **Rina wants to grow vegetables, but her soil is very sandy. Suggest what she could do and explain why.** [2]

Mix in compost or manure: it adds nutrients and humus that holds water, so the soil stays moist.

**3** **Earthworms come up to the surface after heavy rain. Suggest why.** [2]

The soil spaces fill with water, so there is less air for them to breathe; they come up to get air.

**4** **Explain two ways earthworms help plants.** [2]

Their tunnels let air and water reach the roots; their casts add nutrients; they mix humus into the soil.

**5** **Why does the soil of the river delta make Bangladesh good for farming?** [2]

Rivers drop fertile silt every year, adding nutrients and new soil.

ANSWER KEY

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LEARN · PLAY · WONDER

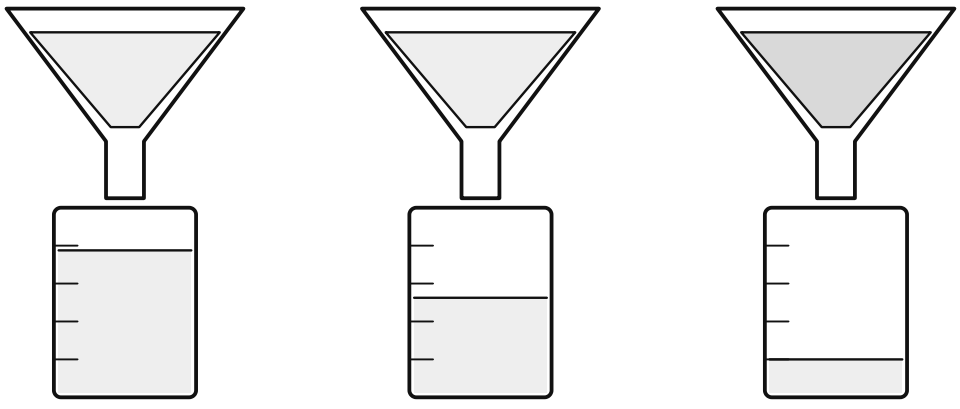
INVESTIGATE · B2 · SOIL

# Which soil holds the most water?

★★☆
Grade 5

**Learning goals**

- Carry out a fair test.
- Present results as a bar chart.
- Use results to answer a real question.



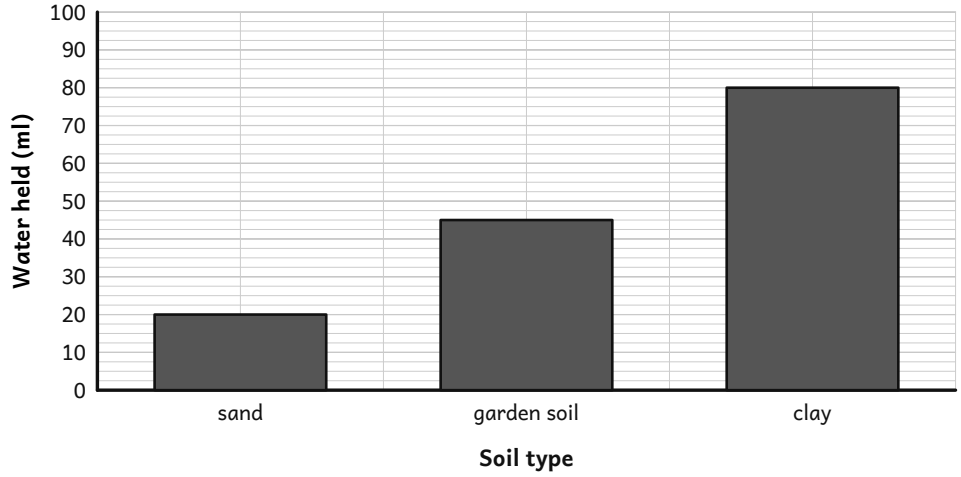

*Equal amounts of soil in each funnel; water poured in at the same time.*

Class 5C put 100 g of each soil in a funnel lined with filter paper. They poured 100 ml of water into each one and measured how much water dripped through in 5 minutes.

**1** **Complete the table to work out how much water each soil held.** [3]

| Soil | Water poured in (ml) | Water that dripped through (ml) | Water held by soil (ml) |
|---|---|---|---|
| sand | 100 | 80 | 20 |
| garden soil | 100 | 55 | 45 |
| clay | 100 | 20 | 80 |

**2** **Draw a bar chart of the water held by each soil.** [4]



**3** **Name three things the class kept the same to make it a fair test.** [3]

mass of soil (100 g), volume of water (100 ml), time (5 minutes), same size funnel and filter paper

**4** **Which soil would be best for a plant that needs a lot of water, and which would be best for a cactus? Explain.** [2]

Clay or garden soil holds water for a thirsty plant; sandy soil drains fast for a cactus, which does not like wet roots.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · B2 · SOIL

# Worms that make money: vermicompost

★★★ Grade 5

**Learning goals**

- Apply soil science to farming.
- Use data to compare.
- Consider benefits and costs.

Science · Economics · Maths · Environment

In many villages of Bangladesh, farmers, many of them women, keep earthworms in concrete rings or large pots filled with cow dung, vegetable peel and straw. The worms eat the waste and turn it into rich, dark **vermicompost** (*kecho sar*) in about 6 to 8 weeks.

Farmers use vermicompost on their own fields and sell the rest. Many say their vegetables grow better and they can use less **chemical fertiliser**. Chemical fertiliser gives quick nutrients, but using too much can harm soil life and wash into rivers and ponds.

Based on vermicompost projects in Bangladesh; numbers below are examples for learning.

**1** **Why does vermicompost help plants grow?** [2]

It is full of nutrients and humus; it holds water and lets air into the soil; it contains helpful soil life.

**2** **Fatema sells vermicompost for 15 taka per kilogram. One worm ring makes 40 kg every 2 months.** [3]

**a)** How much money does one ring earn in 2 months?

40 × 15 = 600 taka

**b)** She has 5 rings. How much do they earn in 2 months?

600 × 5 = 3000 taka

**c)** How much in one year (6 lots of 2 months)?

3000 × 6 = 18 000 taka

**3** **A farmer tested two plots of tomatoes (example results).** [3]

| Plot | Fertiliser | Tomatoes harvested (kg) |
|---|---|---|
| A | chemical fertiliser only | 120 |
| B | half chemical + vermicompost | 135 |

**a)** Which plot grew more tomatoes and by how much?

Plot B, by 15 kg

**b)** Is one test enough to be sure? Explain.

No: the plots might differ in other ways (water, sun, pests); it should be repeated over more plots and seasons.

**4** **Give two benefits of vermicompost for the environment.** [2]

It recycles waste (dung, peel) instead of dumping it; less chemical fertiliser washes into rivers; healthier soil life.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REVIEW AND CHALLENGE · B2 · SOIL

# Show what you know

★★★
Grade 5

**Learning goals**

- Use key vocabulary.
- Answer test-style questions.

**1** **Find the key words.**

| L | U | K | B | P | C | H | A | L | K |
|---|---|---|---|---|---|---|---|---|---|
| I | N | M | G | E | C | C | M | P | D |
| W | S | J | V | P | O | W | O | R | M |
| V | N | O | A | L | M | G | L | F | R |
| O | S | O | I | C | P | D | Y | G | F |
| D | I | U | A | L | O | M | H | H | V |
| C | L | G | I | T | S | S | U | O | N |
| L | T | K | P | C | T | A | M | I | H |
| A | F | D | F | M | O | N | U | E | K |
| Y | R | O | C | K | E | D | S | T | I |

**chalk** **clay** **compost** **humus** **rock** **sand** **silt** **soil** **worm**

**2** **Which soil is sticky when wet and cracks when dry?** [1]

- ○ sandy
- ● **clay**
- ○ chalky
- ○ woodland

**3** **Which soil would fizz with vinegar?** [1]

- ○ sandy
- ○ clay
- ● **chalky**
- ○ garden topsoil

**4** **Design a "worm hotel" to keep earthworms healthy for one week in the classroom. Draw and label it.** [3]

My worm hotel

A clear container with moist, loose soil and some dead leaves; air holes; kept dark and cool; not too wet; worms returned afterwards.

**5** **Challenge. Soil takes hundreds of years to form but can be washed away in one flood. Why should we protect it?** [2]

We need soil to grow almost all our food; it forms very slowly, so lost soil cannot quickly be replaced. Protect it with plants, trees and compost.